

Qatar-1b

R-0598

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_231028 · created 2026-07-25T02:51:42+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460245.6744 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.056
Transit depth (Rp/Rs)^2	3.4 +/- 2.06 [%]
Orbital Inclination (inc)	82.04 +/- 2.88
Airmass coefficient 1 (a1)	0.75 +/- 0.11
Airmass coefficient 2 (a2)	0.2 +/- 0.12
Scatter in the residuals of the lightcurve fit is	14.8 %
Best Comparison Star	#1 - [338, 110]
Optimal Aperture	4.06
Optimal Annulus	5.75
Transit Duration (day)	0.052 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.056 vs 0.1463 (deviation 0.62σ)
Duration fit vs published	0.052 d vs 0.0692 d (deviation 0.59σ)
Mid-time O–C	3.2 min
Depth SNR	3.0
Residual scatter	14.8 %

Light curve

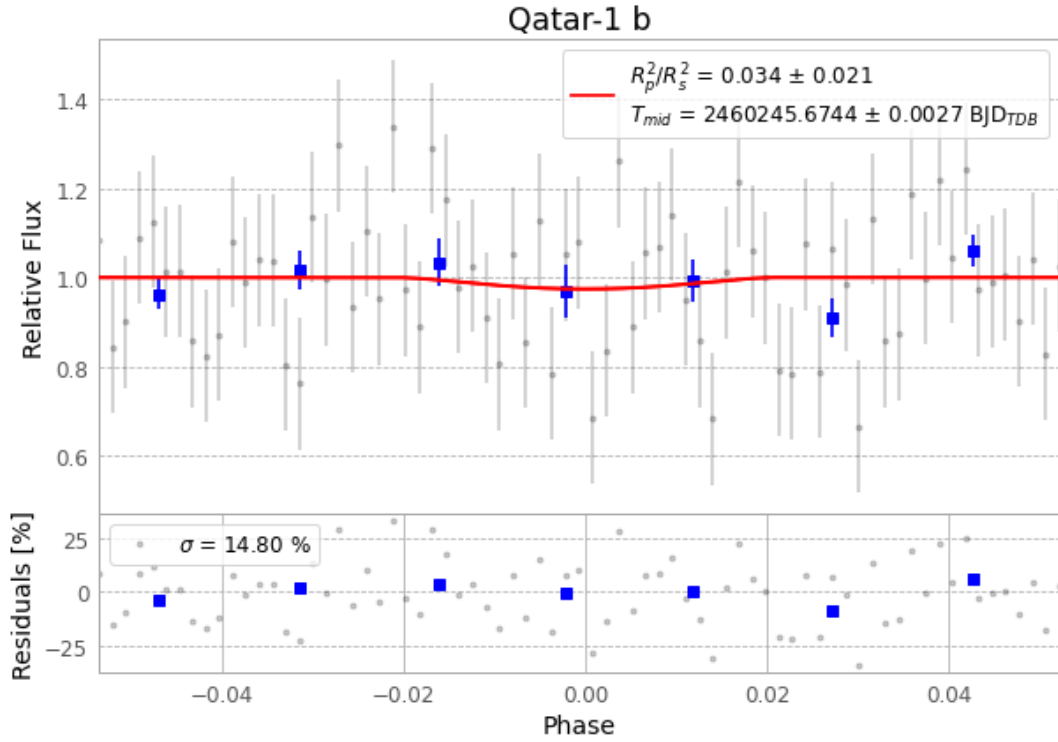


Figure 1 — FinalLightCurve_Qatar-1 b_2023-10-27.png (SHA-256 810165ffd91e9f7e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [114, 401], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ce0804f1567c4805...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

73 FITS frames (48 MB), Qatar-1231028021815.FITS → Qatar-1231028055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-231028.log (2a1c71ea81f3...), FinalLightCurve_Qatar-1 b_2023-10-27.png (810165ffd91e...), FinalLightCurve_Qatar-1 b_2023-10-27.csv (65a6581639e9...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 02:51Z.